INCH-POUND

MIL-G-53133
30 August 1993

MILITARY SPECIFICATION

GENERATOR SETS, SKID MOUNTED, TACTICAL, QUIET, DIESEL ENGINE DRIVEN, ALTERNATING CURRENT 5 THROUGH 60 KILOWATTS

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for generator sets (hereinafter referred to as "set" or "sets"), skid mounted, tactical, quiet, alternating current, diesel engine driven, 5 through 60 kilowatts.

1.2 Classification. The sets shall be tactical and of the following modes and sizes:

MODE:
- I - 50/60 Hz
- II - 400 Hz
- III - 60 Hz

SIZE:
- 5 - 5 kW, MODE II and III
- 10 - 10 kW, MODE II and III
- 15 - 15 kW, MODE I and II (12.5 kW at 50 Hz)
- 30 - 30 kW, MODE I and II (25 kW at 50 Hz)
- 60 - 60 kW, MODE I and II (50 kW at 50 Hz)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Belvoir Research, Development, and Engineering Center, ATTN: SATBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6115

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

1.3 Set rating. The set shall be rated 0.8 power factor (pf), lagging, and shall be re-connectable for the following voltages:

- 5/10 kW, 60 Hz and 400 Hz

  a. 120-volt, single phase, 2 wire

  b. 120/240-volt, single phase, 3 wire

  c. 120/208-volt, 3-phase, 4 wire

- 15/30/60 kW, 50/60 Hz and 400 Hz

  a. 120/208-volt, 3-phase, 4 wire

  b. 240/416-volt, 3-phase, 4 wire

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

VV-F-800	-	Fuel Oil; Diesel.

MILITARY

MIL-L-2104	-	Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-T-5624	-	Turbine Fuels, Aviation, Grades JP-4 and JP-5.
MIL-G-10924	-	Grease, Automotive and Artillery.
MIL-A-11755	-	Antifreeze, Arctic-type.
MIL-A-46153	-	Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-F-46162	-	Fuel, Diesel, Referee Grade.
MIL-L-46167	-	Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-A-53009	-	Additive, Antifreeze Extender, Liquid Cooling Systems.
MIL-T-83133	-	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

(See Supplement 1 for list of associated specification sheets).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Table for Inspection by Attributes.
MIL-STD-209	-	Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	-	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-705	-	Generator Sets, Engine-Driven, Methods of Tests and Instructions.
MIL-STD-810	-	Environmental Test Methods & Engineering Guidelines.
MIL-STD-1400	-	Engine, Gasoline or Diesel, Methods of Test.
MIL-STD-1474	-	Noise Limits for Army Materiel (Metric).

HANDBOOKS

MILITARY

MIL-HDBK-705	-	Generator Sets, Electrical, Measurements and Instrumentation.

(Unless otherwise indicated, copies of federal and military specifications and standards are available form the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DETAILED TEST PLAN FOR FIRST ARTICLE PREPRODUCTION TEST (FAPT) OF TACTICAL QUIET GENERATOR SETS, Nuclear Effects Directorate, US Army White Sands Missile Range, dated December 1989.

(Requests for this document must be referred to: Commander, US Army Belvoir Research, Development, and Engineering Center, ATTN: SATBE-FGM, Fort Belvoir, VA 22060-5606.)

US Army Nuclear and Chemical Agency Memorandum, dated 17 April 1987, Subject: NUCLEAR SURVIVABILITY CRITERIA FOR THE COMMERCIAL GENERATOR SET AND ASSEMBLAGES.

(Requests for this document must be referred to: US Army Nuclear and Chemical Agency, ATTN: MONA-NU, 7500 Backlick Road, Building 5073, Springfield, VA 22150-3198.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

2.2 Non-Government publications. The following document form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the issue of the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington D.C. 20037.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail requirements for individual set size, and mode. The individual set requirements shall be as specified herein and in accordance with the applicable specification sheets (see 6.2). In the event of any conflict between the requirements of this specification and the specifications sheets, the latter shall govern.

3.1.1 Drawings. The drawings forming a part of this specification as listed in the applicable specification sheets, are end product drawings. No deviation from the prescribed drawing requirements is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the set. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.1.2 Size and weight. Set size and operational weight (see 6.4.15) shall be as specified in the applicable specification sheets.

3.1.3 Reliability. Set reliability shall be as specified in the applicable specification sheets.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

Delete: "INSPECTION EQUIPMENT DRAWINGS

3.2 First article (preproduction model). One generator with excitation system and five complete assembled sets of each size and mode shall be furnished for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that the contractor is capable of producing sets that comply with the requirements of this specification. Examination and tests shall be as specified in section 4; and, unless otherwise specified (see 6.2), all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government.

3.3 First article (production model). When specified (see 6.2) five or more sets per size, per mode, shall be furnished for inspection as specified in section 4.

3.4 Materials. Materials shall be in accordance with the drawings listed in the applicable specification sheets.

3.5 Design. The detail requirements are as specified in 3.1.

3.5.1 Engine. The engine, including all systems and components shall meet the requirements as specified herein.

3.5.2 Cranking system. The cranking system shall be 24 volts direct current (DC), negative ground. The set shall not be damaged (see 6.4.3) in the event polarity of the battery cables is reversed.

3.5.3 Batteries. Batteries shall be furnished with electrolyte, in all first article sets. When specified (see 6.2), batteries with shall be furnished with all production sets.

3.5.4 Starting aid. Ether tanks shall be furnished in the first article sets. Production sets shall be furnished without ether tanks.

3.5.5 Governing system. The governing system shall provide the frequency performance as specified in the applicable specification sheets. Any loss of input signal to the governing system shall not cause the set to operate at more than 115 percent of the rated speed.

3.5.6 Generator and excitation system. The generator and excitation system (exciter, voltage regulator, and other accessories necessary to control the output voltage) shall meet the requirements as specified herein and on the applicable specification sheets.

3.5.7 Generator. No damage (see 6.4.3) shall result from running the set at its low idle speed setting for a period of one hour with the excitation system energized and the generator set temperature stabilized in an ambient of 49 °C (120 °F). Phase sequence shall be ABC and L1 to L2 and L3. The generator shall be in electrical and mechanical balance at all speeds up to 125 percent rated speed. The generator shall withstand operation at 2,500 rpm for 15 minutes without damage (see 6.4.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

3.5.8 Temperature rise. Allowable temperature rise of generator coils, windings and mechanical parts, when the generator and auxiliary items are installed in the set, shall not exceed the following:

Coils, windings and connections:

- 105 C° (189 F°) (measured by rise in resistance method)

Generator bearings:

- 50 C° (90 F°)

Poles, cores, and other mechanical devices:

- Same as for the parts in contact with insulation in which the parts are in contact.

3.5.9 Insulation resistance. Insulation resistance of generator windings shall not be less than 1 megohm in an ambient temperature of 25 °C (77 °F).

3.5.10 Dielectric strength. Generator windings shall withstand the following 60 Hz voltages applied for 1 minute:

a. Generator field and exciter windings - 10 times ceiling voltage, but neither less than 1,500 nor more than 3,500 volts (applied between windings and ground)

b. Windings energized by the 24 - volt dc control, cranking and battery charging systems - 500 volts (applied between windings and ground)

c. Generator stator windings - 1,832 volts between phase windings and 1,480 volts between phase windings and ground

d. All others - twice rated voltage plus 1,000 volts (applied between winding and ground and between windings where applicable).

3.5.11 Short circuit. The generator, voltage regulator and excitation system operating as a unit shall withstand two consecutive single-phase line-to-line, single-phase line-to-neutral, and symmetrical 3-phase short-circuits at 5-minute intervals at generator output terminals when operating at rated load (see 6.4.6) and frequency without reduction in dielectric strength to a point where it does not meet the requirements in 3.5.10. The sustained short-circuit current shall not be less than 300 percent of rated output current. 3-phase short circuits shall be for 10 seconds and single phase short circuits shall be for 5 seconds.

3.5.12 Winding resistance. The resistance of each generator winding shall be as specified on the applicable drawings. All resistance values shall be corrected to a temperature of 25 °C (77 °F).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

3.5.13 Saturation curves. The curves listed herein shall be plotted with the generator armature terminal volts on the vertical axis versus the exciter field current on the horizontal axis:

a. Open circuit saturation curve
b. Synchronous impedance curve (short circuit saturation curve)
c. Zero power factor saturation curve
d. Rated load current saturation curve

The exciter field current shall be as specified in the specification sheets.

3.5.14 Phase balance (voltage). The maximum difference in the 3 line-to-neutral voltages shall not be more than 1 percent of rated line-to-neutral voltage. The maximum difference between the voltages of the 120-volt windings of any one phase shall not be more than 1 volt.

3.5.15 Transient reactance, rated voltage. For sets 15 kW and above, the direct axis transient reactance shall not be more than 18 percent for the 3-phase 60 Hz operation of the mode I sets and shall not be more than 16 percent for the 3-phase 400 Hz operation of the mode II sets. For sets 5 and 10 kW, the direct axis transient reactance shall not be more 25 percent for the 120-volt single phase connection and shall not be more than 18 percent for the 3-phase connection.

3.5.16 Negative sequence impedance. For sets 15 kW and above, the generator negative sequence impedance shall not be more than 18 percent for the 60 Hz rating of the mode I sets and shall not be more than 15 percent for the mode II sets. For sets 5 and 10 kW, the generator negative sequence impedance shall not be more than 18 percent with the set connected in the 120/208-volt 3-phase connection.

3.5.17 Flexural vibrations and critical speeds. The sets shall be free from dangerous flexural vibration (see 6.4.4) and dangerous torsional critical speeds (see 6.4.7) between the minimum low idle speed and 115 percent of rated speed.

3.5.18 Overspeed. The 15, 30, and 60 kW sets shall operate at 115 percent of rated speed for 5 minutes without damage (see 6.4.3). Rated speed of the sets shall be that required for the sets to produce rated frequency. The 5 and 10 kW sets shall operate at wide open throttle for 5 minutes.

3.5.19 Instruments, controls, and other devices. Instruments, controls, and devices shall meet all requirements herein when mounted in the set with the set operating under all conditions, including storage, as specified herein.

3.5.20 Protective system. The set shall be equipped with the protective devices specified herein and the applicable specification sheets. In addition to functions specified herein, activation of these devices shall cause the appropriate lamp to illuminate on the malfunction indicator panel.

3.5.20.1 Overspeed device. This device shall activate at 2,200 ±40 rpm and shut down the engine. This device is not required on the 5 and 10 kW sets.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

3.5.20.2 High temperature device. This device shall activate at 107 ±3 °C (225 ±5 °F) - except 104 ±2 °C (220 ±3.5 °F) for 15 kW sets - and shut down the engine before damage.

3.5.20.3 Low oil pressure device. The low oil pressure protective device shall activate at 15 ±3 pounds per square inch (psi) and shut down the engine.

3.5.20.4 Low fuel level device. This device shall activate when the fuel level falls to a point at which operation of the set at rated load is four minutes or more and shut down the set.

3.5.20.5 Overvoltage device. This device shall activate in not more than 1.25 seconds after the voltage in a 120-volt generator coil winding has risen to and remained at any value greater than 153 ±3 volts for not less than 200 milliseconds and shut down the set.

3.5.20.6 Undervoltage device. This device shall be connected across the same generator coil winding as the over-voltage protective device. When the voltage drops below 48 volts this device shall activate instantaneously. In addition, the device shall activate in 6 ±2 seconds after the voltage has fallen to and remained at not more than 99 ±4 volts. The device shall open the set circuit interrupter upon activation. This device is not required on 5 and 10 kW sets.

3.5.20.7 Short-circuit device. This device shall activate within 50 milliseconds in the event set output current in any phase exceeds 425 ±25 percent for all voltage connections. The device shall open the set circuit interrupter upon activation.

3.5.20.8 Overload device. This device shall not activate when current in all three phases is less than 110 percent of rated value. It shall activate within 8 ±2 minutes for 130 percent of rated current in any phase. At currents between 110 percent and 350 percent, the device shall operate on an inverse time principle with the time to trip reduced in proportion to the increase in current. The device shall open the set circuit interrupter upon activation.

3.5.20.9 Reverse power device. This device shall activate if power flow into the set exceeds 20 ±3 percent of rated value. This device is not required on 5 and 10 kW sets. The device shall open the set circuit interrupter upon activation.

3.5.21 Instrument accuracy. The maximum error (see 6.4.12) range of each metering system shall be as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

a. 5 and 10 kW

INSTRUMENT	Maximum Allowable Error Range
Voltmeter	
115-125 volt range	± 9 V
200-250 volt range	± 9 V
Ammeter (AC)	
25 percent Load Current	15-35 percent Rated Current
50 percent Load Current	40-60 percent Rated Current
75 percent Load Current	65-85 percent Rated Current
100 percent Load Current	90-110 percent Rated Current
Frequency meter	
Mode I, 55-65 Hz range	± 1.5 Hz
Mode II, 380-420 Hz range	± 6 Hz

b. 15, 30 and 60 kW

INSTRUMENT	Maximum Allowable Error Range
Voltmeter	
115-125 volt range	± 5 V
200-250 volt range	± 10 V
Ammeter (AC)	
25 percent Load Current	17-33 percent Rated Current
50 percent Load Current	42-58 percent Rated Current
75 percent Load Current	67-83 percent Rated Current
100 percent Load Current	92-108 percent Rated Current
Frequency meter	
Mode I, 55-65 Hz range	± 0.6 Hz
Mode I, 48-52 Hz range	± 0.6 Hz
Mode II, 380-420 Hz range	± 4 Hz
Kilowattmeter	
25 percent Load	12-38 percent Load
50 percent Load	37-63 percent Load
75 percent Load	62-88 percent Load
100 percent Load	87-113 percent Load

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

3.5.22 Malfunction indicator. The malfunction indicator system shall cause the appropriate indicating lamp to energize under activation of the protective device (see 3.5.20).

3.5.23 Lifting and tiedown provisions. Lifting and tiedown provisions shall be provided in accordance with MIL-STD-209. The set, with fuel tank 1/4 full to full, shall remain within 15 degrees from level when lifted.

3.5.24 Interchangeability. All mode sets of one size shall use the same engine, governing system, skid base, housing, frame, AC and DC wiring harnesses, AC and DC schematic diagrams, mounts, batteries, battery charging system, fuel system, air filter, oil filter, fuel filter, muffler, and as many other common components.

3.6 Transportability. The sets shall not be damaged by rough handling (see 6.4.3.2) which could be encountered during rail, truck, trailer, aircraft, or helicopter transportation (see 6.4.8, 6.4.9, and 6.4.10).

3.7 Performance. The set performance shall be as specified herein and in the applicable specification sheets.

3.7.1 Starting, operating and stopping.

3.7.1.1 Starting. The set shall start (see 6.4.1) within 5 minutes under the following conditions:

a. Any individual or combination of the environmental conditions specified in 3.9.
b. With the base of the set in planes from level up to 15 degrees from level.
c. With fuels, lubricants, and coolants as specified herein.

3.7.1.2 Operating. Immediately after starting and with a maximum warm-up period of 5 minutes, the set shall operate without damage (see 6.4.3) or failure (see 6.4.11) under the following conditions:

a. Conditions specified in 3.7.1.1.
b. All loads, continuous and intermittent, up to and including rated load as specified herein and in the applicable specification sheets.

3.7.1.3 Stopping. The set shall stop (see 6.4.2) within 30 seconds or within the time interval specified herein after activation of any device intended to stop the set.

3.7.2 Parallel operation. The requirement for parallel operation is as specified in the applicable specification sheet (see 3.1).

3.7.3 Maximum power. The minimum acceptable maximum power shall be 110 percent of rated load under all operating conditions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

3.7.4 Fuels. The sets, while operating on referee grade fuels conforming to MIL-F-46162; DF-1, DF-2, or DF-A diesel fuel conforming to VV-F-800; JP-5 turbine fuel conforming to MIL-T-5624; or JP-8 turbine fuel conforming to MIL-T-83133 shall meet all requirements herein. The sets, while operating on JP-4 turbine fuel conforming to MIL-T-5624 (with a cetane rating of 30 to 35) shall meet all requirements herein, except rated load may be reduced 15 percent and a maximum of 300 hours of operation on JP-4 per 3,000 hours of set operation shall be required. Operation on JP-4 shall only be required at temperatures -18 to 38 °C (0 to 100 °F) and at altitudes up to 914 meter (m) (3,000 feet).

3.7.5 Lubricants. The engine shall operate on lubricating oil conforming to MIL-L-2104 when temperatures are above -26 °C (-15 °F). The engine shall operate on lubricating oil conforming to MIL-L-46167 when temperatures are at or below 4 °C (40 °F). Grease shall conform to MIL-G-10924.

3.7.6 Coolant. The set shall operate with the following coolants:

a. MIL-A-11755 antifreeze from -32 to 49 °C (-25 to 120 °F) ambient.
b. Water with MIL-A-46153 inhibited antifreeze from -32 to 49 °C (-25 to 120 °F) ambient.
c. Water with MIL-A-53009 additive from 4 to 49 °C (40 to 120 °F) ambient.

3.8 Audio noise. Audio noise sound-pressure levels (SPL) emanating from the set shall not exceed 70 dBA at 7 m (22.9 feet) from the perimeter of the set with all doors closed, when measured at 1.2 m (3.9 feet) above the ground. In addition, audio noise emanating from the set shall not exceed 85 dBA at the operator's station, defined to be 0.7 m (2.3 feet) from the control panel, while the control panel door(s) are open. These requirements apply under all operating conditions specified herein with the set operating at all loads from the no-load to rated load.

3.9 Environmental requirements.

3.9.1 Starting and operating. The set shall start and operate under each and any combination of the following operating conditions:

a. All possible relative humidities, with ambient temperatures ranging from -32 to 49 °C (-25 to 120 °F) at sea level; and from -32 to 35 °C at 1,219 m (-25 to 95 °F at 4,000 feet). The high temperature requirement remains at 35 °C (95 °F) at altitudes above 1,219 m (4,000 feet).

b. Altitudes up to and including 2,438 m (8,000 feet). Rated load may be reduced by 3.5 percent for each 305 m (1,000 feet) above 1,219 m (4,000 feet) (e.g., at 8,000 feet rated load may be reduced by 14 percent).

c. With 127 millimeters (mm) (5 inches) of rain per hour impinging on the set at angles from the vertical up to 45 degrees.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

d. With 355 British thermal units (BTUs) per square foot per hour of solar radiation.

e. With sand and dust particle concentration of up to 1,400 milligrams per cubic meter (mg/m^3). Particle sizes shall range from less than 74 micrometers in diameter to 1,000 micrometers with the bulk of the particles ranging in size from 74 to 350 micrometers.

f. With a steady windspeed of 22 meters per second (m/s) (73 feet per second [ft/s]) and gusts up to 29 m/s (95 ft/s) at a height of 3 m (10 feet) above ground level.

g. With accumulations of ice glaze, freezing rain and hoarfrost of up to 13 mm (0.5 inch) and up to a specific gravity of 0.9.

h. In a salt fog or sea spray environment.

3.9.2 Storage. The set without packaging shall not be damaged (see 6.4.3.1) by exposure to:

a. Storage at -51 to 71 °C (-60 to 160 °F).
b. Salt fog environment.
c. All possible relative humidities.

3.10 High altitude electromagnetic pulse (HAEMP). All sets operating as a single unit and 15 through 60 kW sets while operating in parallel (see 3.7.2) shall withstand the HAEMP nuclear environment as specified in the US Army Nuclear and Chemical Agency Memorandum, dated 17 April 1987, Subject: Nuclear Survivability Criteria for the Commercial Generator Set and Assemblages. The set shall meet the operational requirements specified herein within 5 minutes after being subjected to the specified HAEMP nuclear environment. The recovery may not include the replacement of piece parts or components.

3.11 Electromagnetic interference. The electromagnetic interference emission and susceptibility characteristics of the set shall not exceed the UM04 limits of class C2 equipment of MIL-STD-461, Part 9.

3.12 Treatment and painting. Treatment and painting shall be in accordance with the drawings listed in the applicable specification sheets. When specified (see 6.2), a camouflage pattern shall be applied to the set.

3.13 Maintainability. The maintenance ratio (see 6.4.16) shall not exceed 0.05.

3.14 Endurance. The set endurance requirements are as specified in the applicable specification sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

3.15 Workmanship. Workmanship shall be of a quality to assure delivery of sets that are free from defects (see 6.4.17) resulting from defective material and incorrect manufacturing or assembly practices.

3.16 Government-furnished property. Each set will be mounted in the Government-furnished power unit/power plant configurations listed below.

Size	Nomenclature
5 kW	AN/MJQ-35
10 kW	AN/MJQ-37
15 kW	PU-801
15 kW	PU-802
30 kW	PU-803
60 kW	PU-805

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection (see 6.4.13) requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or repeat any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsible for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Disassembly inspection. Failure of any inspection of the preproduction model generator with excitation system, or sets, shall be cause for disassembly, in the presence of a Government representative, to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings. Parts not complying with requirements shall be rejected and shall be cause for rejection of

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:13Z
Check the source to verify that this is the currentversion before use.

the preproduction model set. Reassembly with acceptable components or parts and reinspection/retest shall be the responsibility of the contractor and at no cost to the Government.

4.1.3 Inspection failure. Should the Government elect to perform or repeat any inspection in this specification, failure of a set to meet any requirement specified herein shall be cause for refusal by the Government to accept production sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on sets previously produced under the contract. Any deficiencies found as the result of such elective inspection will be considered prima facie evidence that all sets accepted prior to the completion of such inspections are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer. The provisions of this paragraph apply, notwithstanding any prior acceptance of preproduction model sets, preproduction model test reports, or production sets. The Government shall notify the contractor of the time and location of any tests the Government elects to perform or repeat. The Government reserves the right to reject the set for not meeting any requirement of this specification even though not performing a test directly related to the specific requirement.

4.1.4 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and inspected/tested in accordance with referenced specifications, standards, and drawings as applicable.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article preproduction inspection (see 4.4).
b. First article production inspection (see 4.5).
c. Quality conformance inspection (see 4.6).
d. Inspection comparison (see 4.9).
e. Inspection of packaging (see 4.10).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in applicable test method document or applicable paragraphs in the specification.

4.3.1 Fuel, lubrications, and coolant for inspection. Referee grade diesel fuel conforming to MIL-F-46162 shall be used for all tests unless otherwise specified herein. DF-1 (VV-F-800) shall be used for all testing with ambient temperatures between -32 to -18 °C (-26 and 0 °F). Lubricating oil shall be as follows:

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:13Z
Check the source to verify that this is the currentversion before use.

LUBRICATING OIL SPECIFICATION	GRADE	DAILY LOW AMBIENT TEMPERATURE (°C/°F)
MIL-L-2104	OE/HDO-15/40	-18 to 49 °C (0 to 120 °F)
MIL-L-2104	OE/HDO-10	-26 to 4 °C (-15 to 40 °F)
MIL-L-2104	OE/HDO-30	-9 to 49 °C (15 to 120 °F)
MIL-L-2104	OE/HDO-40	-4 to 49 °C (25 to 120 °F)
MIL-L-46167	OEA	-32 to 4 °C (-25 to 40 °F)

Coolant shall conform to either MIL-A-11755, water with MIL-A-46153 inhibited antifreeze, or water with MIL-A-53009 additive for all testing conducted at temperatures from 4 to 49 °C (40 to 120 °F). Below 4 °C (40 °F), either MIL-A-11755 or water with MIL-A-46153 inhibited antifreeze shall be used for testing.

4.4 First article preproduction inspection. The preproduction generators, with excitation system, and sets shall be examined and tested as specified herein to determine compliance with this specification and the applicable specification sheets (see 3.1). Noncompliance shall be cause for performing the disassembly inspection specified in 4.1.2.

4.4.1. Examination of components and subassemblies. Examination of components and subassemblies shall be made prior to assembly of the generator or set. Evidence that any components or subassemblies do not comply with the requirements of this specification shall be cause for rejection of that component or subassembly.

4.4.1.1 Examination of generator, excitation system, and sets. Examination of the generator with excitation system and the three sets shall be made without disassembly. Evidence that the generator, excitation system, or sets do not comply with the drawings shall be cause for rejection of the generator, excitation system, or sets.

4.4.2 Preproduction tests. Two preproduction sets shall be subjected to the applicable tests marked "X" under column 1 of table I except for the reliability test. Three preproduction sets shall be subjected only to reliability test of table I. All generators shall be subjected to the generator only tests marked "X" in column 1 of table I. The generator with excitation system shall be subjected to the generator with excitation system tests marked "X" under column 1 of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

table I. In addition to any test specified as part of the preproduction tests, the Government reserves that right to conduct any and all preproduction tests and failures of such additional tests shall have the same effect as failures of those tests specified as preproduction tests.

TABLE I. Test Schedule.

1	2	3	4	5	6	7
First Article		Quality Control				
PPT SEE 4.4	PT SEE 4.5	INDI SEE 4.6.1	SAMP SEE 4.6.2	TEST	TEST METHOD (MIL-STD-705) AND TEST PARAGRAPH (SEE NOTE 1)	BASIC REQUIREMENT PARAGRAPH
				GENERATOR ONLY:		
X		X		High Potential	TM 302.1 (Isolate diodes)	3.5.10
X		X		Insulation Resistance	TM 301.1 (Isolate diodes)	3.5.9
X		X		Winding Resistance	TM 401.1	3.5.12
				GENERATOR WITH EXCITATION SYSTEM:		
X	X			Direct-axis Transient Reactance	TM 425.1	3.5.15
X			X	Generator Power Input	TM 415.1 (Sample at rated load only)	3.1
X	X			Negative Sequence Reactance and Impedance	TM 422.1	3.5.16
X	X			Open Circuit Saturation Curve	TM 410.1 (See Note 2)	3.5.13
X	X			Overspeed (Generator only)	TM 505.3	3.5.18
X		X		Rated Load Current Saturation Curve	TM 413.1 (See Note 2; One point at rated voltage only for individual tests)	3.5.13
X			X	Short Circuit (Mechanical strength)	TM 625.1 (Apply single phase short circuits at each line-to-line and each line-to-neutral connection. Also apply symmetrical 3 phase short circuits at the output terminals. Single phase short circuits are for 5 seconds, 3 phase short circuits are for 10 seconds. Power input to the voltage regulator to sustain the required short circuits may be obtained from a separate source.)	3.5.11

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:13Z
Check the source to verify that this is the currentversion before use.

TABLE I. Test Schedule. (continued)

1	2	3	4	5	6	7
First Article		Quality Control				
PPT SEE 4.4	PT SEE 4.5	IND I SEE 4.6.1	SAMP SEE 4.6.2	TEST	TEST METHOD (MIL-STD-705) AND TEST PARAGRAPH (SEE NOTE 1)	BASIC REQUIREMENT PARAGRAPH
X	X			Synchronous Impedance Curve (Short-circuit Saturation Curve)	TM 411.1 (See Note 2)	3.5.13
X	X			Zero Power Factor Saturation Curve	TM 412.1 (See Note 2)	3.5.13
				GENERATOR SET:		
X	X		X	Noise level	4.8.2.1	3.8
X		X		Circuit Interrupter (Overload Current)	TM 512.2	3.5.20.8
X		X		Circuit Interrupter (Over-voltage and Undervoltage)	TM 512.3	3.5.20.5, 3.5.20.6
X		X		Circuit Interrupter (Short Circuit)	TM 512.1 (Use oscillograph for preproduction qualification model sets only)	3.5.20.7
X		X		DC Control	TM 655.1 (Perform at minimum and maximum settings of the battery charging voltage adjustment control.)	3.1
			X	Sample Endurance	TM 690.1, 4.8.2.7	3.1
X		X		Frequency Adjustment Range	TM 511.2 (At no-load and rated load; See Note 3)	3.1
X		X		Frequency & Voltage Regulation, Stability and Transient Response (Short Term)	TM 608.1 (See Note 2 and 4)	3.1
X	X			Frequency & Voltage Stability (Long Term)	TM 608.2 (See Note 3)	3.1

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:13Z
Check the source to verify that this is the currentversion before use.

TABLE I. Test Schedule. (continued)

1	2	3	4	5	6	7
First Article		Quality Control				
PPT SEE 4.4	PT SEE 4.5	IND I SEE 4.6.1	SAMP SEE 4.6.2	TEST	TEST METHOD (MIL-STD-705) AND TEST PARAGRAPH (SEE NOTE 1)	BASIC REQUIREMENT PARAGRAPH
X	X		X	High Temperature at 49 °C (120 °F)	TM 710.1 at 49 °C (120 °F) (Take temperature rise measurements on the stator windings and exciter field under the following conditions: (See Note 5) All single and 3 phase connections at the maximum and minimum voltage adjustment range specified on the applicable specification sheets. (Also perform TM 619.1 and 619.2 at rated frequency and 3 phase volt-connections.)	3.1, 3.9.1
X		X		High Temperature Protective Device	TM 515.2	3.5.20.2
X	X			Humidity	TM 711.1, 4.8.2.20	3.9
X		X		Indicating Instrument (Electrical)	TM 513.1 (At all rated voltage connections.)	3.5.21
X				Lifting and Tiedown	TM 740.4	3.5.23
X		X		Low Fuel Protective Device	TM 515.5 (See Note 6)	3.5.20.4
X		X		Low Oil Pressure Protective Device	TM 515.1	3.5.20.3
X		X		Malfunction Indicator Exam	4.8.2.11	3.5.22
X	X			Maximum Power	TM 640.1	3.7.3, 3.7.4
		X		Maximum Power	TM 640.4	3.7.3, 3.7.4
X		X		Overspeed (Set)	TM 505.1 (At 115 percent rated speed for 5 minutes)	3.5.18
X		X		Overspeed Protective Device	TM 505.2	3.5.20.1
X		X		Parallel Operation	TM 630.1, 4.8.2.2 (See Note 3)	3.1, 3.7.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

TABLE I. Test Schedule. (continued)

1	2	3	4	5	6	7
First Article		Quality Control				
PPT SEE 4.4	PT SEE 4.5	INDI SEE 4.6.1	SAMP SEE 4.6.2	TEST	TEST METHOD (MIL-STD-705) AND TEST PARAGRAPH (SEE NOTE 1)	BASIC REQUIREMENT PARAGRAPH
X		X		Phase Balance (Voltage)	TM 508.1 (To be measured at load terminals. Also, measure voltages for each winding using separate excitation. On individual production sets measure line-to-line and line-to-neutral at load terminals only)	3.5.14
X		X		Phase Sequence (Rotation)	TM 507.1	3.5.7
X	X			Railroad Impact	TM 740.5, 4.8.2.3	3.6
		X		Regulator Range	TM 511.1 (See note 6)	3.1
X	X			Reliability	TM 695.1, 4.8.2.18.1	3.1
X	X		X	Reverse Battery Polarity	TM 516.5	3.5.3
X		X		Reverse Power Protective Device	TM 516.2 (15 kW size and above only)	3.5.20.9
X		X		Start and Stop	TM 503.1	3.7.1, 3.9
X	X			Voltage Dip and Rise for Rated Load	TM 619.2 (See Note 4)	3.1
X	X		X	Voltage Dip for Low Power Factor Load	TM 619.1 (See Note 4)	3.1
X	X		X	Voltage Modulation	TM 602.1 (At all rated voltage connections.)	3.1
X	X			Voltage Unbalance with Unbalanced Load	TM 620.2	3.1
X	X		X	Voltage Waveform (Oscillographic, Harmonic Analysis)	TM 601.1 and 601.4 (See Note 2; At all rated voltage connections and the following loading conditions: a. No-load b. Rated kW at 0.8 pf lagging c. Rated kW at 1.0 pf)	3.1

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:13Z
Check the source to verify that this is the currentversion before use.

NOTES:

1. Tests shall be conducted with the 240/416-volt connection for sets 15 kW and larger, unless otherwise indicated. For Mode I sets, tests shall be conducted at 60 Hz only unless otherwise indicated. Tests shall be conducted with the 120/208-volt connection for sets 10 kW and smaller, unless otherwise indicated. The Government reserves the right to reject the set for not meeting any requirement of this specification even through not performing a test directly related to the specific requirement.

2. For Mode I sets, test shall be conducted at 50 and 60 Hz at all voltage connections.

3. For Mode I sets, test shall be conducted at 50 and 60 Hz.

4. Test to be conducted at all voltage connections for 5 and 10 kW size sets.

5. For sample tests, only make the runs which gave highest stator and excitor field temperature rises during the first article inspection.

6. For the low fuel protective device test, the requirements of fuel system shall be verified as applicable.

7. Also view waveform with an oscilloscope having a bandwidth of DC to 1.5 MHz and a usable viewing screen of 8 by 10 cm. The oscilloscope gain shall be adjusted such that one cycle of voltage covers approximately the entire viewing screen. For sample tests, only TM 601.4 need to be used for the voltage waveform test.

4.4.3 Inspection failure. Failure of a first article set to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the first article sets.

4.5 First article production inspection. Two production sets shall be subjected to the applicable tests marked "X" under column 2 of table I except for the reliability test. Three production sets shall be subjected only to reliability test of table I. All generators shall be subjected to the generator only tests marked "X" in column 2 of table I. The generator with excitation system shall be subjected to the generator with excitation system tests marked "X" under column 2 of table I. In addition to any test specified as part of the first article production inspections, the Government reserves that right to conduct any and all other test contained in this specification as part of the first article production inspection and failure of such addition tests shall have the same effect as failure of those tests specified as first article production inspections.

4.5.1 Inspection failure. Failure of a production set to meet any requirement specified herein during and as a result of the examination and tests specified in 4.5 shall be cause for rejection of the production sets and shall be cause for

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

refusal by the Government to continue acceptance of production sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on sets previously accepted and produced under the contract. Any deficiencies found as a result of the production inspection will be considered prima facie evidence that all sets accepted prior to the completion of production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.6 Quality conformance inspection

4.6.1 Individual set inspection. Each production set shall be examined and tested (see 4.6.1.1), without disassembly, for conformance to this specification and the drawings (see 3.1). Noncompliance with the drawings shall be cause for rejection.

4.6.1.1 Tests. Each set shall be subjected to the tests marked "X" under column 3 of table I. For generator only tests, the tests shall be conducted prior to assembly into the set. Failure of any test shall be cause for rejection.

4.6.2 Sample inspection. Sampling shall be in accordance with MIL-STD-105 after all sets comprising a lot have passed the inspection specified in 4.6.1. Unless otherwise specified (see 6.2), lot sizes shall consist of not more than 50 sets except that the first lot shall contain not less than 25 sets. Sample size shall be determined using tables I and II-A of MIL-STD-105, special inspection level S3, zero defects are allowable. No adjustment or substitution of components shall be made on the sets selected for sample inspection.

4.6.2.1 Tests. Sets selected in accordance with 4.6.2 shall be subjected to the tests marked "X" under column 4 of table I in the order listed with the exception that the sample endurance test shall be conducted on 50 percent of the samples from each lot or on 2 samples per lot whichever is greater. When this 50 percent results in fraction of sets, the next highest number representing complete sets shall be used. A sample set which fails to meet a requirement during any test shall be considered a defective set. Failure of any sample set to meet any requirement herein shall be cause for rejection of the lot. Rejection shall be the cause for Government to refuse further acceptance of production sets until Government-approved correction of deficiencies has been implemented. The Government may choose that the rejected lot, after correction of deficiencies, undergo any or all of the table I, column 4 tests, prior to acceptance of the lot or acceptance of further production sets. Correction of deficiencies and retesting of rejected sets or lots shall be accomplished by the contractor under Government supervision at no cost to the Government.

4.7 Test schedules. The test schedules shall be as shown in table I. Nonconformance to the applicable requirement paragraph shown in column 7 shall constitute failure of the test and shall be cause for rejection of the generator, the generator with excitation system, or the set, as applicable. The Government

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

reserves the right to reject the equipment for not meeting any requirement herein, even though not performing a test directly related to the specific requirement. The requirements in section 3 shall apply to all tests performed as a part of another test. If maintenance shutdown, as permitted in appendix B, Maintenance Schedule, would be due to occur, because of hours accrued during previous testing, after the beginning of any test, the shutdown shall be postponed until completion of the test or until completion of 100 hours of operation, whichever is lesser.

4.7.1 Table I. Unless otherwise specified (see 6.2), the test schedule shall be as shown in table I.

4.7.2 Table II. When changes to components or materials specified on the drawings are proposed (see 6.3) by the contractor, the contracting officer shall determine which, if any, additional tests listed in table II shall be conducted by the contractor to determine set compliance with the requirements specified herein. All quality assurance provisions of section 4 shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

TABLE II. Additional test schedule.

1	2	3
TEST	TEST METHOD (MIL-STD-705) And TEST PARAGRAPH (SEE NOTE)	BASIC REQUIREMENT PARAGRAPH
	GENERATOR SET:	
Altitude Operation	Before and after REM test. TM 720.1 at 1219 m, 35 °C; and 2438 m, 35 °C (4,000 ft, 95 °F; and 8,000 ft, 95 °F)	3.7.4, 3.9
Drop (Ends)	TM 740.3	3.6
Electromagnetic Interference	4.8.2.5 (At all rated voltage connections at the following load conditions: a. No load b. Rated kW load at 1.0 pf c. Rated kW load at 0.8 pf lagging	3.11
Extreme Cold Storage	TM 731.1, 4.8.2.17	3.9.2
Fuel Consumption	TM 670.1 (At 25, 50, 75, and 100 percent rated load)	3.1
HAEMP	4.8.2.8	3.10
Ice Glaze and Wind	4.8.2.15	3.9.1
Inclined Operation	TM 660.1 (Perform at maximum fuel levels. Perform at full and add positions on the oil dipstick)	3.7.1.1
JP-4 and JP-5 Fuel	4.8.2.12	3.7.4
Maintainability Demonstration	4.8.2.22	3.13
Motor Starting	4.8.2.6	3.1
Rain	4.8.2.17	3.9.1
Reliability, Endurance, Maintainability	4.8.2.18.2	3.1, 3.13
Road	4.8.2.9	3.6
Salt Fog	4.8.2.10	3.9.1
Sand and Dust	4.8.2.14	3.9.1
Size and Weight	4.8.25	3.1
Solar Radiation	4.8.23	3.9.1.d
Starting and Operating (Moderate Cold Battery Start)	TM 701.2 at -32 °C (-25 °F)	3.9.1
Voltage and Frequency Drift	4.8.2.13	3.1
Torsiographing	TM 504.2 (At no load and rated load. Also perform search for dangerous flexural vibration and dangerous torsional critical speeds between low idle speed and 115 percent of rated speed.)	3.5.17

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:13Z
Check the source to verify that this is the currentversion before use.

ficial comments (recommendations, additions, deletions) and any pertinent data

NOTE: Tests shall be conducted with the 240/416-volt connection for sets 15 kW and larger, unless otherwise indicated. For Mode I sets, tests shall be conducted at 60 Hz only unless otherwise indicated. Tests shall be conducted with the 120/208-volt connection for sets 10 kW and smaller, unless otherwise indicated.

4.8 Test procedures. Tests shall be conducted in accordance with MIL-STD-705, MIL-HDBK-705, MIL-STD-1400 and as specified herein. All test instruments shall be of the laboratory type and shall have been calibrated within 30 days of start of testing and calibrated at maximum intervals of 6 months unless otherwise specified by the instrument manufacturer. Written certification from the instrument manufacturer is required if other than a 6 month calibration interval is used. Instruments shall have a valid calibration during their period of use. Instruments used in calibration shall be at least five times the accuracy of the instrument being calibrated. Calibrated reference instruments of lesser accuracy than standard and which are not used for any other purpose may be used for the required periodic check of test instruments. Instruments shall be calibrated at the frequencies at which they are going to be used. Direct-reading instruments shall have a least 0.5 percent accuracy and shall be connected to indicate the most accurate portion of their range. On DC instruments, the readings shall not be made on the lower 15 percent of the scale. On AC instruments, the readings shall not be made on the lower one-third of the scale. When the test methods call for the use of voltage and frequency recording-type meters, other than oscillograph, the Texas Instrument Company Model PDRHXFHXVA-A16-XT or the Gould part number 2108-2202-005542 or Gould part number 30-V8202-106383 shall be used. Oscillograph galvanometer frequency response shall be flat (±5 percent) from DC to not less than 3,000 Hz. When recording meters are specified for any part of a test, turn on the recording meters prior to starting the warm-up period of the set and record continuously for the duration of the test. Unless otherwise specified, the recording meters shall be operated at a minimum speed of 2.5 mm/min (6 in/hr) during the portions of the test where steady state loading conditions exist and shall be operated at a minimum speed of 2.5 mm/s (6 in/min) at least 30 seconds before, during, and after a load change. Individual tests requiring a series of data points shall be repeated if an interruption occurs during the test (e.g., data points being taken for plotting a curve, etc.). Exhaust smoke testing shall be in accordance with the method specified in MIL-STD-1400. Exhaust smoke conditions shall be measured during all altitude testing.

4.8.1 Government performed tests. The maintainability demonstration, road, and HAEMP tests will be accomplished by the Government at Government expense at a Government installation.

4.8.2 Tests.

4.8.2.1 Audio noise test. Instrumentation and procedures for the audio noise test shall conform to MIL-STD-1474. Measure audio noise sound pressure level at no-load and rated load. The microphone(s) shall be located at 7 m (22.9 feet) from the perimeter of the set and 1.2 m above the ground. Measure audio noise

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

sound pressure levels at 0.7 m from the control panel, at a height equal to the control panel, with the set on the ground, and with the control panel door(s) open.

4.8.2.2 Parallel operation test. For first article preproduction model testing (column 1 of table I), two sets of each size and mode shall be tested in accordance with method 630.1 of MIL-STD-705. For individual set quality assurance testing (column 3 of table I), each production set shall be tested in parallel with one of the first production lot sets (see 6.2) randomly selected by the Government, in accordance with method 630.1 of MIL-STD-705 with the following changes:

a. Control panel instrumentation on the sets shall be used.
b. Load connections shall be no-load, half-load, and rated load.
c. Where 1 hour of operation is referenced, substitute 15 minutes of operation.

4.8.2.3 Railroad impact test. Test shall be performed in accordance with TM 740.5 of MIL-STD-705, with the following conditions:

a. Two sets shall be used.
b. The two sets shall be mounted in opposite direction to each other.
c. The sets shall be mounted in accordance with set rail tiedown procedures.
d. The sets shall not require any type of packaging for the test.

4.8.2.4 Drop test. The drop test shall be performed in accordance with TM 740.3 of MIL-STD-705 at a height of 9 inches.

4.8.2.5 Electromagnetic interference test. The electromagnetic interference test shall be performed in accordance with MIL-STD-462 and Appendix A.

4.8.2.6 Motor starting test. The motor starting test shall be performed by use of a motor rated NEMA Code F, in accordance with MG-1. The motor shall be loaded with a flywheel or equivalent having an inertia equal to that of the motor rotor. Satisfactory starting is defined as acceleration of the motor to rated speed without tripping any generator set protective devices.

4.8.2.7 Sample endurance test. Sample sets shall be operated 250 hours per set for the 5 and 10 kW sizes and 300 hours per set for the 15-60 kW sizes, in accordance with MIL-STD-705, TM 690.1. The disassembly portion contained in TM 690.1 is not required. The number of operating hours at each load shall be proportional to the hours specified in the TM 690.1 cyclic load schedule. JP-8 fuel per MIL-T-83133 shall be used for this test. The sets shall be fueled utilizing the auxiliary fuel system on the set. The set fuel tank shall be inspected for leaks during the test. The sample sets shall complete the test with zero defects. Defects are defined in 6.4.17.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

4.8.2.8 HAEMP tests and analyses. This test shall be conducted in accordance with the Detailed Test Plan for First Article Preproduction Test (FAPT) of Tactical Quiet (TQ) Generator Sets and Towed Assemblages, Nuclear Effects Directorate, U.S. Army Test and Evaluation Command, dated December 1989.

4.8.2.9 Road test. Each configuration shall undergo 10 road cycles as defined in 6.4.9 on the Government-furnished property listed in 3.16. The set shall be in operating condition and all fluids shall be at normal level, except the fuel tank shall be one-half full. Each set shall be towed by an appropriate vehicle. TM 608.1 of MIL-STD-705, shall be conducted at the beginning of the test and at the end of every two road cycles. These tests shall be conducted at rated load only. Prior to assembly and at the conclusion of the road test, the generator set shall be subjected to an audio noise test (see 4.8.21).

4.8.2.10 Salt fog test. The salt fog test shall be performed in accordance with method 509.2 of MIL-STD-810. Two sets shall be subjected to two 48-hour cycles as follows:

24 hours - salt fog exposure
24 hours - standard ambient (drying)

Salt concentration shall be a 5 ±1 percent solution. The sets shall be tested in their normal operating mode. TM 608.1 of MIL-STD-705 shall be performed within 4 hours before the start of the salt fog cycling. After performing TM 608.1, but prior to the start of the salt fog cycling, isolate the exciter field and measure their insulation resistances in accordance with TM 301.1 of MIL-STD-705, except that the values need not be corrected for temperature. After completion of the salt fog cycling, but prior to the post-cycling TM 608.1 test, measure the insulation resistance of all circuits initially measured in accordance with TM 301.1, except that the set shall not be operated prior to this test and measured values shall not be corrected for temperature. Upon completion of the post-cycling TM 301.1 test, reconnect all circuits and perform TM 608.1 within 4 hours after removal of the set from the test chamber. The set shall be examined for corrosion or other physical damage (see 6.4.3) resulting from this test.

4.8.2.11 Malfunction indicator exam. Indicator operation shall be noted throughout all testing.

4.8.2.12 JP-4 and JP-5 fuel test. The fuel test shall be performed using JP-4 fuel as follows:

a. Perform a high temperature test at 38 °C (100 °F) in accordance with TM-710.1 of MIL-STD-705, except that section 710.1.3.2 shall be modified as follows: Perform only paragraphs a and g (608.1 only).

b. Perform a moderate cold test at 18 °C (0 °F) in accordance with TM 701.2 of MIL-STD-705, except that Section 701.2.3.2k shall be modified as follows: Perform 608.1 only.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:13Z
Check the source to verify that this is the currentversion before use.

c. Operate each test set for a total of 300 hours in accordance with TM 690.1 of MIL-STD-705.

The above procedures shall be repeated using JP-5 except high temperature test shall be performed at 49 °C (120 °F). Acceptance criteria for the 300-hour operation run shall be zero relevant or critical failures. JP-4 shall conform to MIL-T-5624 with a cetane rating of 30-35, and JP-5 shall conform to MIL-T-5624.

4.8.2.13 Voltage and frequency drift test. Follow the same procedure in TM 608.2 of MIL-STD-705, except that the set is initially stabilized at an ambient temperature which differs by 33 C° (60 F°) from the final stabilization temperature and the test is conducted for 8 hours. This test may be accomplished as the environmental chamber is being warmed from a cold temperature test. Temperature change shall be less than 5.5 C° per hour (10 F° per hour).

4.8.2.14 Sand and dust test. The sand and dust test shall be conducted in accordance with method 510.2 of MIL-STD-810, Procedure II, and with the following conditions:

a. Perform TM 608.1 of MIL-STD-705.

b. Test shall be performed at prevailing ambient temperatures and relative humidities.

c. Air velocity shall be maintained at a minimum of 32.2 km/h (20 MPH).

d. Sand composition shall be as described in paragraph 1-3.2.d(2), method 510.2 of MIL-STD-810.

e. Sand concentration shall be 1.4 g/m^3.

f. The test shall consist of four 90-minute intervals. The set shall be oriented with each side exposed to the blowing sand for one 90-minute interval. The set shall be operated at no load for a minimum of 10 minutes during the last half of each 90-minute interval.

g. Following completion of the four 90-minute intervals, TM 608.1 of MIL-STD-705 shall be performed.

4.8.2.15 Ice glaze and wind test. The set shall be subjected to the combined action of wind and ice as follows:

a. Perform TM 608.1 of MIL-STD-705.

b. With the ambient temperature between -23 and 0 °C (-10 and 32 °F), and with a steady windspeed of at least 22.25 m/s (73 ft/s), deliver a uniform rainspray on the non operating set until 12.7 mm (0.5 inch) of ice glaze has accumulated on the top of the set.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

c. After accumulation of 12.7 mm (0.5 inch) on the top of the set, the set shall be started and operated at no load for a minimum of 1 hour.

d. After 1 hour of operation at no load, TM 608.1 of MIL-STD-705 shall be performed.

4.8.2.16 Extreme cold storage test. The test shall be conducted in accordance with TM 731.1 of MIL-STD-705 except at an extreme cold temperature of -51 °C (-60 °F) instead of -62 °C (-80 °F).

4.8.2.17 Rain test. Conduct test in accordance with Appendix C at 5 inches of rain per hour.

4.8.2.18 Reliability, endurance, maintainability test. The reliability test shall be conducted on 3 sets per mode, in accordance with MIL-STD-705, TM 695.1. The disassembly portion contained in TM 695.1 is not required. The number of operating hours at each load shall be proportional to the hours specified in the TM 695.1 cyclic load schedule. The sets shall be fueled utilizing the auxiliary fuel system on the set. The set fuel tank shall be inspected for leaks during the test.

4.8.2.18.1 Reliability test. The reliability test shall be conducted using table III. The reliability failure criteria shall be in accordance with table III. All failures shall be in accordance with 6.4.11. JP-8 fuel per MIL-T-83133 shall be used for this test. One critical failure (see 6.4.11.d) shall constitute a failure of the reliability test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

TABLE III. Reliability test schedule.

Set Size	Mode	Relevant Failures Allowed	Test Hrs per Set	Total Hours
5 kW	60 Hz	5	1167	3500
5 kW	400 Hz	5	1113	3813
10 kW	60 Hz	4	1345	4034
10 kW	400 Hz	5	1235	3652
15 kW	50/60 Hz	4	1205	3616
15 kW	400 Hz	5	900	2700
30 kW	50/60 Hz	4	1345	4034
30 kW	400 Hz	5	1344	4032
60 kW	50/60 Hz	5	1286	3858
60 kW	400 Hz	5	951	2854

4.8.2.18.2 Reliability, endurance, and maintainability test. When specified, the reliability, endurance, and maintainability test shall be conducted using table IV. The reliability failure criteria shall be in accordance with table IV. All failures shall be in accordance with 6.4.11. One critical failure (see 6.4.11.d) shall constitute failure of the endurance test. JP-8 fuel per MIL-T-83133 shall be used for the half of the test and referee grade diesel per MIL-F-46162 shall be used for the other half of the test. Each set shall run an additional amount of hours to accumulate 6,000 hours per set. Maintenance ratio shall be determined during the reliability portion of the test.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:13Z
Check the source to verify that this is the current version before use.

TABLE IV. Reliability, endurance, and maintainability test schedule.

Set Size	Mode	Relevant Failures Allowed	Test Hrs per Set	Total Reliability Hours
5 kW	60 Hz	17	3145	9440
5 kW	400 Hz	17	3016	9047
10 kW	60 Hz	12	3179	9538
10 kW	400 Hz	16	3132	9396
15 kW	50/60 Hz	13	3051	9153
15 kW	400 Hz	22	3023	9068
30 kW	50/60 Hz	12	3179	9538
30 kW	400 Hz	14	3081	9243
60 kW	50/60 Hz	15	3129	9386
60 kW	400 Hz	21	3107	9320

4.8.2.19 Hot storage test. Perform in accordance with MIL-STD-705, test method 732.1 at 71 °C (160 °F).

4.8.2.20 Humidity test. The humidity test shall consist of 5 consecutive 48-hour cycles as specified in MIL-STD-705, test method 711.1.

4.8.2.21 Smoke test. Exhaust smoke testing shall be in accordance with the method specified in MIL-STD-1400. Exhaust smoke condition shall be measured during altitude testing at 1,219 and 2,438 m (4,000 and 8,000 feet).

4.8.2.22 Maintainability demonstration. The maintainability demonstration shall consist of performance of all scheduled maintenance (PMCS and 250/300 hour servicing) per Appendix B. Arctic mittens and MOPP IV gear shall be used. Any item of service which can not be performed with arctic mittens or MOPP IV gear shall constitute a failure. Item 12, tighten hardware and replace insulation (noise) shall be excluded.

4.8.2.23 Solar radiation test. The solar radiation test shall be in accordance with Appendix D.

4.9 Inspection comparison. The Government may select sets at any time during the contract production period and subject these sets to any examination and test specified herein, necessary to determine that the selected sets meet all

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:13Z
Check the source to verify that this is the current version before use.

requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Sets will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and production sets. Acceptance of an inspection comparison set shall not exclude the remaining sets from the quality conformance inspection and acceptance provisions specified in section 4. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.9.1 Inspection failure. Failure of an inspection comparison set to meet any requirement specified herein, during and as a result of the examination and tests specified in 4.9, shall be cause for refusal by the Government to continue acceptance of production sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on sets previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all sets accepted prior to completion of the inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.10 Inspection of packaging. The inspection shall include the examination of the packaging presented against the packaging instructions listed on drawings in the applicable specification sheets.

4.10.1 First article preproduction packaging inspection. This inspection shall be performed on one complete pack per generator set size for each of the packaging levels specified (see 5.1). One or more packaging defects (see 6.4.18) shall be cause for rejection.

4.10.2 Quality conformance packaging inspection. This inspection shall be performed on each individual set for each of the packaging levels specified (see 5.1). One or more packaging defects (see 6.4.18) shall be cause for rejection.

5. PACKAGING

5.1 Preservation, packing, and marking. Each complete generator set shall be preserved, packed and marked in accordance with the packaging drawings listed in the applicable specification sheets. The contracting officer shall provide the level of packaging and preservation (see 6.2).

5.2 Connection tag. A connection tag shall be attached to one of the load terminals, stating: "This set is connected for 120/208 volts". All sets shall be shipped in the 120/208 V connection. All mode I sets shall be shipped in the 60 Hz configuration.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:13Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.

6.1 Intended use. The sets are intended to supply power for multipurpose use in military applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.2 and 2.2).

c. Associated specification sheets, number and date (see 3.1).

d. Time frame required for submission of first article preproduction model (see 3.2).

e. When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests. The contracting officer should specify which examination and tests, will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).

f. When initial production inspection is required and number of sets to be furnished, when applicable (see 3.3).

g. Whether batteries are to be furnished with production sets (see 3.5.3).

h. Which paint color code is to be used for camouflage pattern (standard, desert, winter/snow) (see 3.12).

i. Government-furnished property (see 3.16).

j. Size of lots for production sets (see 4.6.2).

k. Delivery and test schedule (see 4.7.1).

l. How production paralleling sets are to be retained and maintained during life of the contract (see 4.8.2.2) and how sets are to be refurbished and delivered at the end of the contract.

m. Level of preservation and packing required (see 5.1).

n. Whether failure reports are required.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:13Z
Check the source to verify that this is the current version before use.

6.3 Configuration management. The generator sets controlled by this specification are under the configuration management authority of the DoD Project Manager for Mobile Electric Power. All engineering change proposals (ECPs), request for waivers (RFWs), and request for deviations (RFDs) must be processed in accordance with DLAR 4120.7, AR700-101, AFR 400-50, NAVMATINST 4120.100A, MC011310.8C, Joint Operating Procedures Management and Standardization of Mobile Electric Power Generating Sources.

6.4 Definitions.

6.4.1 Start. A set is considered to have started when it is operating at rated voltage and speed without the further use of starting aids.

6.4.2 Stop. A set is considered to have stopped when all rotating members are at zero rpm, with the exception of a turbo charger if used.

6.4.3 Damage. Damage is defined as any failure (see 6.4.11), temperature or humidity related (see 6.4.3.1), rough handling damage (see 6.4.3.2) or degradation in life. The blowing (opening) of a replaceable fuse is not considered damage, provided it is performing its intended function.

6.4.3.1 Temperature and humidity damage. Temperature and humidity damage is defined as conditions causing malfunction of any component or part, corrosion, breakage, deformation, or reduction of insulation resistance below 50,000 ohms.

6.4.3.2 Rough handling damage. Rough handling damage is defined as any condition resulting in malfunctioning of the set, liquid leakage, deformation, loosening, breakage, or change of fit of any component or part.

6.4.4 Dangerous flexural vibration. Dangerous flexural vibration is defined as a vibration which occurs at a speed at which maximum stress in the shaft from flexural vibration exceeds 9,000 psi.

6.4.5 Stable (engine) operating conditions. Stable (engine) operating conditions are the conditions specified for short-term, steady-state performance.

6.4.6 Rated load. Rated load is rated kilowatt at rated power factor, rated frequency and rated voltage.

6.4.7 Dangerous torsional critical speed. Dangerous torsional critical speed is defined as the speed at which maximum vibrating stress in the shaft from torsional vibration exceed 5,000 psi.

6.4.8 Railroad transportation. Railroad transportation shall be interpreted to mean impact speeds up to and including 12.88 km/h (8 MPH) under test conditions specified in 4.8.2.3.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:13Z
Check the source to verify that this is the current version before use.

6.4.9 Truck or trailer transportation. Truck or trailer transportation is defined as the conditions encountered during four cycles of a road test, each cycle consisting of the following, with the set mounted on an Army truck or trailer:

ROAD CONDITION	DISTANCE km (Mi)	SPEED km/h (MPH)
Paved Highway	161 (100)	up to 80.5 (50)
Level Cross-country	161 (100)	up to 32.2 (20)
Hilly Cross-country	80.5 (50)	up to 32.2 (20)
Belgian Block	9.7 (6)	up to 32.2 (20)

6.4.10 Aircraft and helicopter transportation. Aircraft and helicopter transportation shall be interpreted to mean a 22.86 cm (9 inch) end drop under test conditions specified in MIL-STD-705, method 740.3.

6.4.11 Failure. A failure is defined as the inability of an item to perform within specified limits. The contracting officer will classify failures into the categories described below. The following definitions are applicable for section 3 requirements and the specification sheets.

a. Relevant failure. Any failure of an item which prevents the set from starting, stopping, meeting the power output requirements, meeting audio noise requirements, any failure which renders any device of the protection system (see 3.5.20) inoperative, or any failure which causes a critical or catastrophic hazard to personnel or equipment. Relevant failure shall be used to calculate the MTBF requirement.

b. Nonrelevant failure. Any failure of an item which is not used to compute reliability. Examples of nonrelevant failures are as follows:

- Secondary failures caused by failures in the powered equipment or other occurrences in the set environment when integral set protection is not provided against such equipment failures of occurrence, e.g., explosion or fire

- Failures due to characteristic of the load, e.g., waveform distortion caused by saturated inductors

- Failures resulting from operating items beyond requirement

- A failure of an item which does not prevent the set from meeting the power output requirements specified herein, e.g., a panel light burns out

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:13Z
Check the source to verify that this is the current version before use.

- Failures due to operator error where procedures are documented in technical manuals, instruction plates mounted on the set, or both, e.g., use of improper lubricant

- Failures due to design deficiencies when subsequent testing demonstrates that the design deficiency has been corrected may be assessed and rescored as nonrelevant if the Government determines testing has proved the design deficiency has been corrected

c. Pattern failure. The occurrence of two or more failures of the same part in the same failure mode constitutes a pattern failure. A pattern failure, whether scored relevant or nonrelevant, may be a cause for rejection of the sets.

d. Critical failure. A critical failure is defined as a relevant failure requiring removal of the engine, cylinder head, oil pan, gear cover, or ac generator to repair.

6.4.12 Accuracy/error. Accuracy is a ratio which defines the limit of error expressed as a percentage of full scale value. Error is the difference between the indication and the true value of the quantity measured. It is the quantity which, when algebraically subtracted from the indication, gives the true value. A positive error denotes that the indication of the meter is greater than the true value.

6.4.13 Inspection/test. Inspection/test is the examination and testing of supplies or services including, when appropriate, raw materials, components, and intermediate assemblies to determine conformance with contract requirements.

6.4.14 First article set. First article sets are defined as preproduction or production models.

6.4.15 Operational weight. The operational weight is the total weight of the set including fuel, lubricating oil, electrolyte, and coolant at full capacities. The weight also includes ground rods and attaching hardware, paralleling cables where applicable, and auxiliary fuel line.

6.4.16 Maintenance ratio. The maintenance ratio is defined as the total maintenance manhours per total operating hours for all scheduled and unscheduled maintenance (excluding preventive maintenance checks and services by the operator) of the set. Maintenance manhours include any and all manhours expended for scheduled and unscheduled manhours (excluding preventive maintenance checks and services by the operator) before, during and after operation, including time expended for inspection, diagnosis, and adjustments of the set and repair of failed components and assemblies.

6.4.17 Defect. A defect is any nonconformance of the set with either the specification or drawings specified in the applicable specification sheets.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

6.4.18 Packaging defect. A packaging defect is defined as a noncompliance with any of the packaging instructions contained on drawings in the applicable specification sheets.

6.5 First article pack. Any changes or deviation of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor's obligation to preserve, pack, and mark the generator sets in accordance with this specification.

6.6 Inspection report. The contractor shall furnish, within the time interval specified (see 6.2), inspection reports in accordance with MIL-STD-831 for preproduction inspection and production inspection. The inspection reports shall also include the following:

a. All test data obtained from applicable tests.

b. Description of malfunction, damages, failures, and adjustments (other than adjustments permitted by this specification) which occur during inspection.

c. Causes and analysis of malfunction, damages, or failures, and reasons for adjustments.

d. Corrective actions taken or required.

6.7 Noise limits. Where the limit of Category D can be documented as being clearly beyond the state-of-the-art, per MIL-STD-1474, 5.1.1.2, selection of another noise limit should be considered by the procuring activity, as described in MIL-STD-1474, acceptance requirements (5.1.1.3). Particular attention should be given to 5.1.1.2.2.d. When permission is granted to exceed the limits of category D, the procuring activity should insure that operator and maintenance manuals are prepared in accordance with MIL-STD-1474, manuals requirement (4.3).

6.8 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD Far Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title
4.8	DI-QCIC-80798	Calibration Certificate of Traceability

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:13Z
Check the source to verify that this is the current version before use.

The above DID was cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.9 Subject term (key word) listing.

Noise
Nuclear
Mobile

6.10 Appendix C procurement document requirements. The following items must be specified in the individual procurement document:

a. The voltage connection and frequency at which this method is to be performed.
b. The voltage and frequency requirements specified in TM 608.1, paragraph 608.1.5a.
c. Any operational checks to be performed at the conclusion of this test, if other than as specified herein.
d. Rainfall in inches per hour, if other than as specified herein.
e. Angle of rain impingement, if other than as specified herein.
f. Velocity of wind, if other than as specified herein.

6.11 Government-furnished property. The contracting officer should arrange to furnish the Government-furnished property listed in 3.16.

6.12 First article. When a first article inspection is required, the item(s) should be a preproduction model and initial production model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Preparing activity:
Army - ME

Project 6115-0264

Review activities:
Navy - MC
DLA - GS

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:13Z
Check the source to verify that this is the current version before use.

APPENDIX A

ELECTROMAGNETIC EMISSION AND SUSCEPTIBILITY REQUIREMENTS FOR CONTROL OF ELECTROMAGNETIC INTERFERENCE

10. SCOPE

10.1 Scope. This appendix is a mandatory part of the specification. The information contained herein is intended for guidance only. This document clarifies the limits, test methods and general procedures to be used for measurement of EMI on mobile electric power generating sources (MEPGS). Clarification as to the use of MIL-STD-462 is included. This document is intended to be used as a clarifying supplement for the use of MIL-STD-462 and does not replace MIL-STD-462.

20. APPLICABLE DOCUMENTS

20.1 Government documents

20.1.1 Standards.

* MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
** MIL-STD-810 - Environmental Test Methods & Engineering Guidelines.

* Reference is to Notice 3 of MIL-STD-462.
** Reference is to MIL-STD-810D.

30. GENERAL REQUIREMENTS

30.1 Class C equipment requirement. MIL-STD-461 establishes Class C2 for engine generators and associated components, Uninterruptible Power Sets (UPS) and Mobile Electric Power (MEP) equipment supplying power to or used in critical areas. The UM04 limits for conducted emissions and radiated emissions listed in Part 9 of MIL-STD-461 are applicable to sets rated at 240 kVA or less. The UM04 limits for radiated emissions may be extended to sets with a rating greater than 240 kVA when specified in the procurement document. The UM04 limits for radiated susceptibility are applicable to all generator sets regardless of rating.

30.2 Limits and frequency ranges. Part 9 of MIL-STD-461 establishes the following information:

a. Conducted emissions, 0.015 to 50 mHz, power leads.
b. Radiated emissions, 14 kHz to 1 gHz, electric field.
c. Radiated susceptibility, 2 mHz to 10 gHz, electrical field.

Figures 9-1 and 9-2 of Part 9 establish the criteria for broadband conducted and radiated emissions. The criteria for radiated susceptibility is that the set

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:13Z
Check the source to verify that this is the currentversion before use.

shall not exhibit any malfunction, degradation of performance, or deviation from specified indications beyond the tolerances indicated in the individual equipment specification when subjected to the following radiated E-fields in the frequency range of 2 mHz to 10 gHz:

FREQUENCY RANGE	E-FIELD (VOLTS/METER)
2 to 400 MHz	10
400 MHz to 10 gHz	5

Above 30 mHz the requirement shall be met for both horizontally and vertically polarized waves.

40. QUALITY CONFORMANCE INSPECTION

40.1 Test procedures. Applicable test procedures for determining conformance to UM04 limits are contained in Notice 3 (EL) to MIL-STD-462. Specifically, methods CE04, RE02, and RS03 shall be used for conducted emissions, rated emissions, and radiated susceptibility, respectively. These test procedures are clarified by the attached supplementary test methods.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:13Z
Check the source to verify that this is the current version before use.

APPENDIX A

TEST METHOD NO. 1
CONDUCTED EMISSIONS

1. Test method. MIL-STD-462, Notice 3 (EL), method CE04, procedure (A) Conducted Emissions 50 kHz to 50 mHz, power leads.

2. Frequency limits. 0.015 to 50 MHz.

3. Applicability. Applicable in testing to requirement of figure 9.1 of part 9 of MIL-STD-461.

4. Apparatus

a. Delete: The use of Line Impedance Stabilization Networks (LISNs) and 50 OHM resistive terminations is not required.

b. Add: Fifty-foot load cable and resistive/reactive load bank. The rating of the load bank shall be of sufficient capacity to load the MEPGS to rated capacity at both unity and 0.8 power factor.

5. Procedure. With the 50-foot cable connected to MEPGS output and energized, determine the broadband conducted emissions (0.015 - 50 mHz) on each conductor at the end of the cable at no-load, rated kW load at 1.0 pf, and rated kW load at 0.8 pf at all voltage and frequency connections.

6. Failure criteria. Conducted emissions exceeding the limit of figure 9.1 of MIL-STD-461 shall be cause for rejection.

7. Other

a. The 50-foot cable shall be energized during the no load test.

b. The 10 microfarad feedthrough capacitors used in superseded test methods are not used in this method.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:13Z
Check the source to verify that this is the current version before use.

APPENDIX A

TEST METHOD NO. 2
RADIATED EMISSIONS

1. Test method. Notice 3 (EL), method RE02, Radiated Emissions, electrical field.

2. Frequency limits. 14 kHz to 1 gHz.

3. Applicability. Applicable in testing to requirements of figure 9.2 of Part 9 of MIL-STD-461.

4. Apparatus. Add a 50-foot load cable and a resistive/reactive load bank. The rating of the load bank shall be of sufficient capacity to load the MEPGS to rated capacity at both unity and 0.8 power factor.

5. Procedure

a. In lieu of continuously welded bond for the counterpoise, bonding straps shall be used.

b. With the 50-foot cable connected to the MEPGS output and energized, determine the radiated emissions (14 kHz - 1 gHz) at no-load, rated kW load at 1.0 pf, and rated kW load at 0.8 pf at all voltage and frequency connections.

6. Failure criteria. Radiated emissions exceeding the limit of figure 9.2 of MIL-STD-461 shall be cause for rejection.

7. Other. The 50-foot cable shall be energized during the no-load test.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:13Z
Check the source to verify that this is the current version before use.

APPENDIX A

TEST METHOD NO. 3
RADIATED SUSCEPTIBILITY

1. Test method. Notice 3 (EL), Method RS03, Radiated Susceptibility, 14 kHz to 12.4 gHz, electrical field.

2. Frequency limits. 2 mHz to 10 gHz.

3. Applicability. Applicable to testing MEPGS.

4. Apparatus. Add a 50-foot load cable and a resistive/reactive load bank. The rating of the load bank shall be of sufficient capacity to load the MEPGS to rated capacity at both unity and 0.8 power factor. The output monitor shall be a recording voltage and frequency meter in accordance with MIL-HDBK-705.

5. Procedure

a. With the 50-foot cable connected to the set and energized, subject the set to electric fields to 10 V/m (2-400 mHz) and 5 V/m (400 mHz to 10 gHz) while operating at no-load, rated kW load at 1.0 pf, and rated load at 0.8 pf and at all voltage and frequency connections.

b. The set output voltage shall be monitored throughout the test with the recording voltage and frequency meter operating at 6 inches per hour until voltage or frequency variations as a function of field strength or frequency are noted. Upon observing such variation, the chart speed shall be increased to 6 inches per minute and the frequency spectrum in question shall be rescanned for acceptance/rejection analysis.

6. Failure criteria. Failure of the output voltage or frequency to remain within the 30-second short-term stability bandwidth (specified in the procurement document) when subjected to the specified electrical fields shall be cause for rejection.

7. Other. The 50-foot cable shall be energized during the no load test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

APPENDIX B

MAINTENANCE SCHEDULE.

10. SCOPE

10.1 Scope. The appendix is a mandatory part of the specification. The information contained herein is intended for compliance only.

20. APPLICABLE DOCUMENTS. This section is not applicable to the appendix.

30. MAINTENANCE SCHEDULE.

#	Interval (hours)	Set Size (kW)	Item to be Inspected	Procedures Check for and have repair or adjust as necessary
1	Daily	5-60	Housing	Check doors, panels, hinges, and latches for damaged, loose or corroded items. Inspect air intake and exhaust grills for debris
2	Daily	5-60	Identification plates	Check to ensure identification plates are secure.
3	Daily	5-60	Skid Base	Inspect skid base for cracks or corrosion
4	Daily	5-60	Acoustical Materials	Ensure that acoustical materials are securely in place and free of damage.
5	Daily	5-60	Engine Assembly	Check for loose, damaged or missing hardware.
6	Daily	5-60	Fuel System	Inspect fuel system for leaks, damaged, loose or missing parts.
7	Daily 250 300	5-60 5-10 15-60	Fuel Filter / Water Separator	Inspect fuel filter / water separator for leaks, proper mounting, cracks damage, or missing parts. Replace fuel water separator element.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

APPENDIX B

MAINTENANCE SCHEDULE. (continued)

#	Interval (hours)	Set Size (kW)	Item to be Inspected	Procedures Check for and have repair or adjust as necessary
8	Daily	30-60	Ether Start System	Check for deteriorated, loose or missing parts.
9	Daily	5-60	Lubrication System	Inspect lubrication system for leaks, damaged, loose or missing parts. Check engine oil level. Check engine oil for contamination.
10	Daily 1500	5-60	Radiator	Check radiator for leaks, damage or missing parts. Drain coolant and flush radiator. Add proper coolant.
11	Daily	5-60	Hoses	Check hoses for leaks or cracks.
12	Daily	30-60	Cooling Fan	Check fan for damage or looseness. Check for unusual noise being emitted from fan area.
13	Daily 1500	5-60	Fan Belt	Inspect belts for cracks, fraying and looseness. Replace fan drive belt(s)
14	Daily	5-60	Over-flow bottle	Check over-flow bottle for leaks or missing parts
15	Daily	5-60	Exhaust System	Check muffler for leaks and exhaust system for corrosion, damaged, or missing parts

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

APPENDIX B

MAINTENANCE SCHEDULE. (continued)

#	Interval (hours)	Set Size (kW)	Item to be Inspected	Procedures Check for and have repair or adjust as necessary
16	Daily	5-60	Air Cleaner Assembly	Inspect air cleaner assembly and piping for loose or damaged connections Check restriction indicator for clogged element.
17	Daily 250 300	5-60 5-10 15-60	Batteries	Check electrolyte level. Test batteries for charge state. Replace if charge state is low.
18	Daily	5-60	Battery cables	Inspect cables and connectors for corrosion, loose, damaged or missing parts
19	Daily	15-60	Output Box Assembly	Check for loose or damaged wiring or cables.
20	Daily	5-60	Indicators and Controls	Check all indicators and controls for damaged or missing parts. Ensure all indicators are operating properly.
21	Daily	5-60	Control Box Harness	Check for loose or damaged wiring.
22	Daily	15-60	Parallel Cable	If required for generator set operation, inspect parallel cable for damage.
23	Daily	5-60	Ground Rod Cable and Connection	Inspect ground rod and cable for loose connections, break, damage and corrosion.
24	250 300	5-10 15-60	Engine Lube Oil	Drain lube oil and add proper lube oil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

APPENDIX B

MAINTENANCE SCHEDULE. (continued)

#	Interval (hours)	Set Size (kW)	Item to be Inspected	Procedures Check for and have repair or adjust as necessary
25	250 300	5-10 15-60	Engine Lube Filter	Remove and replace engine lube oil filter.
26	300	15-60	Governor Linkage	Clean and lubricate governor actuator linkage rod ends.
27	1,000 1,500 3,000	5 10 15-60	Engine Fuel Injectors	Replace engine fuel injectors.
28	3,000	15-60	Engine Valve Tappets	Adjust engine valve tappets.
29	250 300	5-10 15-60	Air Filter	Replace air filter element.
30	300	30-60	Crankcase Breather	Replace crankcase breather.
31	250 300	5-10 15-60	Hardware and Sound Insulation	Tighten hardware and check placement of insulation. Replace insulation as required.
32	1,500	5-60	Interior of set	Clean radiator, exterior surface, and engine compartment.
33	1,500	5-60	Magnetic Pick-up	Remove, clean, and reinstall magnetic pick-up.
34	1,500	15-60	Governor Actuator	Replace governor actuator.
35	3,000	5-10	Governor Systems	Examine all interior and exterior components and renew as necessary.
36	3,000	5-10	Muffler	Check for leaks, restriction, accumulation of carbon and replace as necessary.
37	3,000	15-60	Fuel Injection Pump	Replace fuel injection pump.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

APPENDIX C

TEMPORARY METHOD
RAIN TEST

10. SCOPE

10.1 Scope. The appendix is a mandatory part of the specificatin. The information contained herein is intended for compliance only.

20. APPLICABLE DOCUMENTS. This section is not applicable to the appendix.

30. General. Since most generator sets are expected to be operated outdoors, without shelter of any kind, this test is performed to assure proper operation during a heavy rain storm.

30.1 Apparatus. Instrumentation for measuring load conditions and rain shall be as described and illustrated in MIL-HDBK-705. In addition, a test area capable of providing the required rainfall and wind conditions shall be provided.

30.1 Procedure.

30.1.1 Preparation for test.

a. Perform test method 608.1 within 4 hours before the start of the rain test.

b. Place the generator set in the rain environmental location with external connections made to simulate field installation conditions as closely as possible.

c. Connect the load instrumentation in accordance with the applicable figure of MIL-HDBK-705, method 205.1, paragraph 205.1.10 for the voltage connection and frequency specified in the procurement document.

d. The test area shall be at normal ambient temperature (68 to 86 °F or 20 to 30 °C) at the beginning of the test and no further regulation of temperature is required.

30.1.2 Test.

a. A simulated rainfall of 8 ±1 inches per hour or as otherwise specified in the procurement document shall be produced by water spray nozzles of such design that the water is emitted in the form of small droplets, rather than a fine mist. The temperature of the water shall be above 40 °F (4.44 °C). The spray nozzles shall be located so that the water drops impinge on the set at angles between 15 degrees and 45 degrees from the vertical. The water shall be dispersed as uniformly as possible over the entire area by means of a wind source producing wind velocities up and including 40 mph

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

(18 m/s). The wind velocity shall be measured at the position of the set prior to placement of the set in the facility.

b. Open the set control panel doors, if provided. Subject the set to the water spray for 3 consecutive hours. Each side of the set shall be exposed to simulated blowing rain for 30 minutes beginning with the control panel end of the set. At the beginning of the last hour of the test, and with the simulated blowing rain on the control panel end of the set, perform TM 608.1. Continue operating the set during the simulated blowing rain until completion of TM 608.1 or completion of the third hours, whichever conditioner is the longer of the two.

c. Immediately after exposure to the simulated blowing rain and at the rain test site, examine the set for evidence of water penetration and damage. After completion of the examination of the set, start the set and perform TM 608.1 at rated load only.

30.2 Results. The data sheet shall indicate the length of test, quantity and incident angle of the water, wind velocity, any malfunction, water penetration, and water damage (see figure 1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

TEST DATA

ITEM 15KW, 60HZ, 120V, SINGLE-PHASE GENERATOR SET
MFGR. ENGENSETS, INC.
MODEL NO. SE-15.0-MD
SERIAL NO. 16039

DEFENSE CONTRACT ADMINISTRATION SERVICE
PHILADELPHIA REGION

RAIN TEST

TEST NO. 16
SHEET 1 OF 1
DATE 2 SEPT., 1970
JOB NO. 0582
PROJ. ENGR. [illegible]
RECORDER/OBSERVER [illegible]

INST.→		116	218	301		516	711	5077									037	1076
READ NO.	TIME	TERM VOLTAGE	LINE CURRENT AMPS	OUTPUT POWER KW	KW	POWER FACTOR	FREQ	RAIN DEPTH									WATER TEMP °F	AVG AMB TEMP °F
UNITS	HRS	VOLTS	AMPS X40	X40	X1		Hz	IN										
SYM.			X1															
COL.	1	2	3	4		7	8	9	10	11	12	13	14	15	16	17		
	0800	—	—	—	—	—	—	0	START RAIN								64	67
	0900	—	—	—	—	—	—	4.1	PLACED SET IN CHAMBER								64	71
	0930	—	—	—	—	—	—	6.3									64	70
	1000	—	—	—	—	—	—	8.2									65	70
	1030	—	—	APPLIED RATED LOAD		.80	60.0	10.1									65	70
	1045	START	SET	.375	15.0	.80	60.0	12.2									65	72
	1100	120.0	3.90	1.56	.375	15.0	.80	60.0	14.3								65	74
	1130	120.0	3.90	1.56	.375	15.0	—	—	16.2	STOPPED RAIN - INSPECTED SET							65	74
	1145	SHUT	DOWN	SET	—	—	—	—										
	1200	—	—	—	—	—	—	—		VISUAL INSPECTION REVEALED MINOR WATER PENETRATION IN GENERATOR AREA OF SET, NO OTHER WATER WAS OBSERVED INSIDE SET								

NOTES:
LINE CURRENT MEASURED USING C.T. No 1211

SIMULATED DATA FOR ILLUSTRATIVE PURPOSES ONLY

FIGURE 1. Typical test record for rain test. X-4668

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

APPENDIX D

SOLAR RADIATION

10. SCOPE

10.1 Scope. The appendix is a mandatory part of the specificatin. The information contained herein is intended for compliance only.

10.2 Objective. To determine if the tactical quiet generator sets can withstand the effect of solar loading without physical damage or compromises in operational ability.

20. APPLICABLE DOCUMENTS. This section is not applicable to the appendix.

30. CRITERIA

a. The set shall not be damaged (see below) or show degradation in operational performance by the effects of solar radiation with up to 355 British Thermal Units (BTUs) per square foot per hour of solar radiation when tested in accordance with MIL-STD-810D, Test Method 505.2.

b. The damage is defined as:

(1) Change in strength, stress, or loss of structural integrity due to differential expansion and contraction of dissimilar materials.
(2) Jamming or loosening of moving parts or linkages.
(3) Loss of seal or gasket integrity.
(4) Deteriorated electronic or electrical components or wiring.
(5) Blistering, fading, cracking, or peeling of paints.
(6) Softening or weakening of glued parts.

40. TEST PROCEDURE

a. The solar radiation test will be conducted on two sets. Before the sets are placed in the solar radiation test chamber the following preparation will be made:

(1) Perform Test method 608.1a, of MIL-STD-705.
(2) Fill the set's fuel tank and all other liquids to rated capacity.
(3) Install thermocouples at the following locations:
    (a) Engine compartment.
    (b) Control cubicle.
    (c) Fuel tank.
    (d) Top, front, rear, and side surface of housing.

b. Each side will singularly be exposed to four 24-hour, hot-dry test cycles described in MIL-STD-810D, Test Method 505.2, Procedure I, paragraph

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

I-3-2.a and I-3-2.b(1). The set will be placed in the chamber and positioned so that one vertical side and top of the set receives the greatest amount of radiation during the first 24-hour cycle so that the equal exposure to the radiation is accomplished to the entire set by the end of the test.

c. After completion of the four cycles, thoroughly examine the set for physical damage.

d. Remove the set from the chamber and perform Test Method 608.1a, of MIL-STD-705.

50. ANALYTICAL PROCEDURE

The results of the solar radiation test will be reviewed and compared to the criteria for compliance. Inability to meet the voltage and frequency criteria specified in the specification and applicable specification sheets, or evidence of physical damage as described above in 30., will constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

specification, as appropriate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-G-53133	2. DOCUMENT DATE (YYMMDD) 930830

3. DOCUMENT TITLE Generator Sets, Skid Mounted, Tactical, Quiet, Diesel Engine Driven, Alternating Current 5 Through 60 Kilowatts General Specification For

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468 (2) AUTOVON 654-3468
c. ADDRESS (Include Zip Code) S Army Belvoir RDE Center ATTN: SATBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:13Z
Check the source to verify that this is the current version before use.